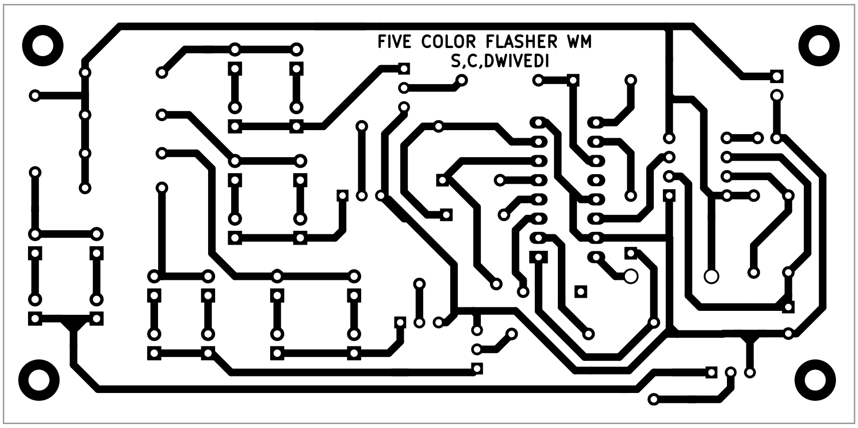

Fig. 3: Actual-size PCB pattern

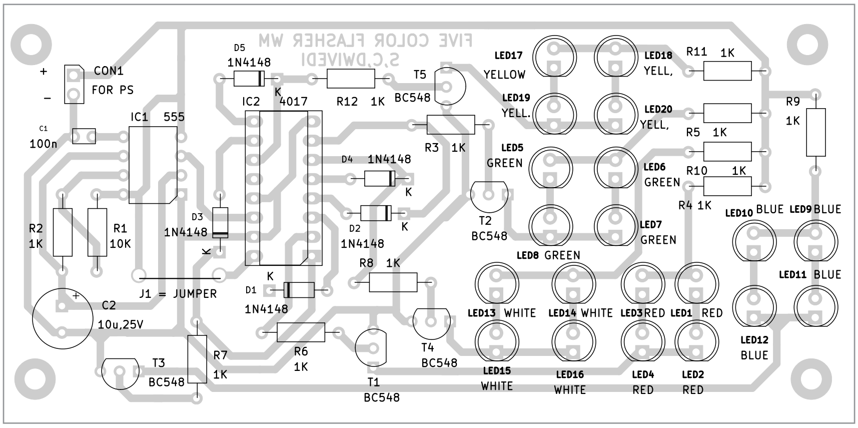

Fig. 4: Component layout of the PCB

other pins of IC 4017 are outputs used to drive the LEDs for flashing.

As mentioned earlier, the 555 timer IC works in astable multivibrator mode. Based on the above-mentioned components, the frequency of the output pulse is around 12Hz.

The output of the 555 timer is supplied as clock input to the 4017 decade counter. This 12Hz frequency can be changed by changing the timing components R1, R2, and C2. In place of R2 you can use a pot to vary the output frequency.

Whenever there is a low to high transition on the clock signal, that is, on the rising edge, the output of the counter is incremented by 1. You cannot change the flash rate in this project as the astable multivibrator built around IC1 generates a fixed frequency.

IC2's five outputs Q0, Q2, Q4, Q6, and Q8 are connected to the anodes of respective 1N4148 diodes D1 though D5. The remaining five outputs Q1, Q3, Q5, Q7, and Q9 are connected to the cathodes of respective 1N4148 diodes.

So, two consecutive outputs drive individual transistors T1 through T5. The bases of all these five transistors are connected to the outputs of 4017 to drive various colour LEDs.

Here, we have used all ten outputs of 4017 to flash the LEDs. Outputs Q0 and Q1 control the red LEDs, outputs Q2 and Q3 control the green LEDs, outputs Q4 and Q5 control the blue LEDs, outputs Q6 and Q7 control the white LEDs, and outputs Q8 and Q9 control the yellow LEDs.

PARTS LIST

Semiconductors:	
IC1	- 555 IC timer
IC2	- 4017 decade counter
D1-D5	- 1N4148 signal diode
LED1-LED4	- 5mm red LED
LED5-LED8	- 5mm green LED
LED9-LED12	- 5mm blue LED
LED13-LED16	- 5mm white LED
LED17-LED20	- 5mm yellow LED
T1-T5	- BC548 npn transistor
Resistors (all 1/4-watt, ±5% carbon):	
R1	- 10-kilo-ohm
R2-R12	- 1-kilo-ohm
Capacitors:	
C1	- 100nF ceramic disk
C2	- 10µF, 25V electrolytic
Miscellaneous:	
CON1	- 2 pin connector
	- 9V or 12V battery
	- Breadboard
	- Jumper wires

The four red LEDs are driven by transistor T1 when outputs Q0 and Q1 of IC2 goes high. Likewise, the four green LEDs are driven by transistor T2 when outputs Q2 and Q3 of IC2 go high. This goes on for the blue, white, and yellow LEDs driven by transistors T3, T4, and T5, respectively.

Working of the circuit is simple. As soon as power is switched on, the 5-colour flasher start flashing. First of all the four red LEDs glow, then the four green LED glow, followed by the four blue, white, and yellow LEDs, each. Thereafter the cycle repeats endlessly.

An actual-size, single-side PCB layout for the 5-colour flasher is shown in Fig. 3 and its component layout in Fig. 4. After assembling the circuit on PCB, enclose it in a suitable box. Fix all the 20 LEDs in a pattern of your liking, like a star or a circle.

CON1 in used to connect the 12V battery to the flasher circuit in a suitable box. If you are using a compact 9V battery, it can be accommodated within the box. EFY

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